

ZOUHOUR BELLAMINE

AI & Software Engineer | Full-Stack | Machine Learning & Generative AI

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PROFILE

Data Engineering and Decision Systems Engineer specialized in Artificial Intelligence and intelligent software development. Experience in Computer Vision, Machine Learning, Generative AI, and full-stack application development. Currently seeking a Junior opportunity in AI Engineering, Software Engineering, Machine Learning, or Data Engineering.

PROJECTS

Spark AI – Generative AI Marketing Automation SaaS

2026 – Present

- Designed and developed a SaaS platform that automatically generates marketing content from product images.
- Integrated Generative AI models to generate product titles, descriptions, social media posts, calls to action, and hashtags.
- Developed a frontend/backend architecture using Next.js and FastAPI, including REST APIs, image processing, and authentication.
- Implemented Supabase for storage, authentication, and data management, with Docker containerization.
- Technologies:** Python, FastAPI, Next.js, React, PostgreSQL, Supabase, LLMs, Prompt Engineering, Docker.

AI Research Copilot – AI-Powered Research Platform

2026

- Developed an intelligent research workspace combining conversational AI, web research, and document analysis.
- Implemented a RAG pipeline for document ingestion, semantic search, and contextual response generation.
- Integrated Gemini and Exa for AI-powered web research with source citations, persistent memory, and PDF/DOCX report export.
- Technologies:** Python, FastAPI, Next.js, React, TypeScript, Gemini, RAG, PostgreSQL, pgvector, Supabase, Exa, Docker.

Text-to-SQL Application with Streamlit

- Developed an application that converts natural language queries into executable SQL queries on a local SQLite database.
- Integrated an LLM for SQL query generation and developed a Flask backend for query processing.
- Built an interactive Streamlit interface for querying the database using natural language.
- Technologies:** Python, Flask, Streamlit, Groq, Llama 3, SQLite.

PROFESSIONAL EXPERIENCE

Final-Year Internship – Computer Vision – Valeo Enfidha

February 2025 – June 2025

- Designed and developed an automated system for detecting visual defects after laser engraving using computer vision techniques.
- Implemented and evaluated a YOLOv8 model for anomaly detection on control buttons, including data annotation and preparation.
- Developed a Flask backend to integrate and deploy the computer vision model.
- Technologies:** Python, YOLOv8, OpenCV, PyTorch, Flask, Roboflow, Computer Vision.

Data Scientist Intern – Zen Group

June 2024 – July 2024

- Developed predictive models to estimate olive oil export quantities using Decision Tree Regressor and Gradient Boosting.
- Developed a Flask backend and React interface to interact with model predictions.
- Technologies:** Python, Scikit-learn, Flask, React, Pandas, NumPy, Machine Learning.

Android Application Development – Tunisie Telecom

June 2023

- Developed a Java Android application for invoice consultation with local data management using SQLite.
- Technologies:** Java, Android Studio, XML, SQLite.

SKILLS

Technical Skills

- Languages:** Python, Java, JavaScript, TypeScript
- Machine Learning:** Scikit-learn, TensorFlow, PyTorch, OpenCV, NLP
- Generative AI:** LLMs, RAG, Prompt Engineering
- Frameworks:** FastAPI, Flask, Next.js, React
- Deep Learning:** CNN, RNN, LSTM
- Databases:** PostgreSQL, Supabase, MongoDB, MySQL, SQLite
- Tools:** Git, GitHub, Docker, JIRA, Agile/Scrum
- Systems:** Windows, Linux

Soft Skills

- Teamwork Adaptability

CERTIFICATIONS

Large Language Models (LLMs) – 365 Data Science

LPIC-1 101-500 – PAQ ENET'Com

Microsoft Azure Fundamentals

Microsoft Azure AI Fundamentals

Microsoft Azure Data Fundamentals

EDUCATION

National School of Electronics and Telecommunications of Sfax (ENET'Com)

2022 – 2025

Engineering Degree in Data Engineering and Decision Systems

Preparatory Institute for Engineering Studies of Monastir

2020 – 2022

Preparatory Degree for Engineering Studies — Rank: 288/1400 (Physics-Chemistry)

LANGUAGES

French: Advanced

English: Advanced

Arabic: Native